American Journal of Primatology 47:209–222 (1999)

BRIEF REPORTS

Effects of Separation and Novelty on Distress Vocalizations and Cortisol in the Common Marmoset (*Callithrix jacchus*)

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In socially-bonding species, separation from familiar attachment figures is widely known to stimulate a physiological and behavioral stress response. This study investigated the hormonal and vocal responses of adult common marmosets to separation from familiar group members and to 24 hr of cohabitation with an unfamiliar opposite-sex conspecific. All subjects were removed from their home cages and placed into a novel environment for 20 min. In one group, marmosets were exposed to an unfamiliar, opposite-sex partner in the novel environment and remained paired with this partner for the 24 hr test period. In three other groups, marmosets experienced the novel environment alone and subsequently were returned to their original social- or single-housing condition, or kept separate from their social groups for a 24 h period. Blood samples were collected the day before, and at 30 min, 90 min, and at 24 h after separation. Cortisol responses were differentially affected by the length of separation and the presence of unfamiliar conspecifics. Brief separation followed by the return to the social group had minimal effect on plasma cortisol levels. All marmosets produced high levels of separation calls in the novel environment, but there was no apparent relationship between calling and cortisol levels. The lack of a temporal relationship between the production of distress vocalizations and serum cortisol has previously been noted in squirrel monkey and rhesus monkey infant separation studies; the behavioral and physiological responses to separation appear to be similarly dissociated in the marmoset. Further, the characteristics of a separation environment can differentially affect the hormonal response by adult marmosets without differentially affecting their behavioral response. Am. J. Primatol. 47:209–222, 1999. Published 1999 Wiley-Liss, Inc.†

Key words: social separation; cortisol; marmosets; separation vocalizations; *Callithrix jacchus*

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Received 14 January 1998; revision accepted 8 September 1998

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INTRODUCTION

In mammalian species exhibiting attachment behavior, involuntary disruption of social bonds invariably stimulates a characteristic separation response, consisting of behavioral agitation, the tendency to seek reunion with the attachment figure, and adrenal activation [Ainsworth, 1972; Bowlby, 1973]. An extensive literature documents the effects of separation from an attachment object in nonhuman primates [see Hennessy, 1997 for review]. For example, studies of mother-infant bonding have found that both infants [Mendoza et al., 1980; Gunnar et al., 1981; Coe et al., 1983; Levine & Wiener, 1988; Wiener et al., 1990] and mothers [Levine et al., 1993; Champoux & Suomi, 1994] exhibit behavioral signs of agitation, such as the production of vocalizations and elevations in cortisol following involuntary separation.

Cortisol concentrations have been clearly linked to the production of alarm calls by rhesus macaques, where mothers treated with metyrapone to suppress cortisol produced significantly fewer calls in response to threats to their infants than did control mothers [Bercovitch et al., 1995]. In the separated rhesus monkey infant, the rate of separation calling, which declines over time, correlates with the magnitude of the cortisol decline [Gunnar et al., 1981]. In contrast, several studies have noted temporal dissociations between the behavioral and the physiological response to involuntary separation, with the behavioral response preceding an elevation in cortisol levels [e.g., Coe et al., 1978; Hennessy et al., 1979; Bayart et al., 1990; Wiener et al., 1990; Levine et al., 1993]. Additionally, behavioral and physiological components of a separation response can be attenuated by the presence of familiar peers [Hennessy et al., 1979; Gunnar et al., 1980; Coe et al., 1982; Hennessy, 1984; Levine & Wiener, 1988]. Other studies have found that removing singly-housed nonhuman primates from a familiar environment and exposing them to a novel environment can both activate the hypothalamic-pituitary-adrenal (HPA) axis and elicit vocal production in infant rhesus monkeys and squirrel monkeys of all ages [Gunnar et al., 1980; Coe et al., 1982; Hennessy, 1984, Levine & Wiener, 1988].

In adults, the response to separation from familiar conspecifics may depend upon the intensity of existing affiliative bonds between group members [Mendoza & Mason, 1986; Mendoza et al., 1991a,b]. In adult rhesus monkeys or squirrel monkeys, separation from familiar social partners generally evokes few physiological signs of separation distress which cannot be accounted for by exposure to a novel environment [Hennessy, 1986; Mendoza & Mason, 1986; Gust et al., 1993, 1994]. In contrast, in the case of the monogamous titi monkey, separation of adult pairmates elevates cortisol levels and elicits vocalizations [Mendoza & Mason, 1986]. Like the titi monkey, common marmosets (*Callithrix jacchus*) normally live in social groups, consisting of a breeding pair and their offspring, and are generally accepted as a pairbonding species [Epple, 1975; Poole, 1978; Evans, 1983; Evans & Poole, 1983, 1984]. Common marmoset social groups are characterized by low levels of intragroup aggression, high levels of affiliation, and aggression against intruders [Epple, 1967; Poole, 1978; Evans, 1983; Evans & Poole, 1984; Anzenberger, 1985, 1988]. Involuntary separation from familiar group members elicits a behavioral separation response, characterized by the prolific production of vocalizations ("phee calls"), in marmosets of all ages [Epple, 1968; Pook, 1978; Hubrecht, 1985; Jones et al., 1993; Norcross & Newman, 1993; Newman, 1995]. Singly-housed adults also produce separation calls after being removed from a familiar environment and placed in a novel environment [Goedeking & Newman, 1987; Jones et al., 1993]. Concentrations of circulating

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and excreted cortisol have recently been established as indicators of social stress in marmosets and tamarins. Separation from mates or other familiar social partners elevates cortisol levels in Wied's black tufted-ear marmosets [Smith & French, 1997] and cotton-top tamarins [Ziegler et al., 1995]. Johnson et al. [1996] found that isolating separated members of established common marmoset pairs for 2 weeks significantly elevated mean cortisol levels, but did not present a temporal analysis of the cortisol changes across this time. Further, the relationship of cortisol levels to the production of separation vocalizations has not been previously investigated in marmosets.

The purpose of the present study was to investigate the temporal relationship of the adult common marmoset's behavioral response, as assessed by the production of separation phee calls, and physiological response, as assessed by serum cortisol levels, to varying lengths of separation from familiar conspecifics. We also examined whether singly-housed animals showed similar behavioral and physiological responses after removal from a familiar environment. Finally, we investigated whether the response to prolonged separation from familiar conspecifics could be attenuated by the presence of an unfamiliar, opposite-sex conspecific, as might be expected during the early stages of new pair formation.

METHODS

Subjects and Housing

The subjects were 35 common marmosets aged 1.1–7 years. Animals were housed either in established social groups of two to five members, consisting either of peers (same-sex) or related animals, or singly (n=4). Groups were housed in standard wire mesh cages (61 × 61 × 152 cm) provided with nest boxes and branches. None of the subjects were members of a reproductive pair; the reproductive status of the females was not hormonally evaluated prior to the experiment. Animals were exposed to a 12:12 light cycle in a colony room maintained between 21–25.5°C, and 55–65% humidity. Further details of colony housing and husbandry can be found in Norcross & Newman [1993].

Test Groups and Study Design

Subjects were assigned to one of four test groups. Each group experienced separation from the home cage and/or familiar social group. Group treatments differed by a) the duration of separation (i.e., 20 min vs. 24 hr) from familiar group members, b) the presence of unfamiliar opposite-sex conspecifics during the separation, or c) the pre-separation social condition (i.e., group-housed vs. singly-housed). We tested the effect of the duration of separation from familiar social partners to determine if the response to brief separations (i.e., 20 min) that may occur in natural settings, such as during feeding, differs from the response to prolonged separations (i.e., 24 hr), such as may occur after death or desertion [e.g., Fernandez-Duque et al., 1997]. We tested the effects of the presence of unfamiliar opposite-sex conspecifics during brief separation to determine if the separation response might be attenuated in this situation, which might be found during the establishment of a new breeding pair [Hennessy et al., 1979; Gunnar et al., 1980; Coe et al., 1982; Hennessy, 1984; Levine & Wiener, 1988]. The purpose of including singly-housed animals as a group was two-fold. First, we wanted to investigate if the physiological separation response in this species results from the social separation from familiar partners [e.g., Mendoza & Ma-

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son, 1986] or if it can be elicited after removal from a familiar environment and exposure to a novel environment [e.g., Gunnar et al., 1980; Coe et al., 1982; Hennessy, 1984, Levine & Wiener, 1988]. Second, we wanted to investigate if a response to removal from a familiar environment was qualitatively similar to that shown by animals separated from familiar social companions.

Colony size, age constraints, and other ongoing experimental protocols limited the numbers of animals of each sex available for inclusion in each group, and necessitated testing each group serially across 2 years. To test for hormonal and behavioral responses to prolonged separation from familiar group members, seven males were removed from familiar group members, placed alone into the novel environment for 20 min, and then kept separate in a novel standard cage for 24 hr ("SEPARATED"; median age 19.2 months, range 17.1–25.6 months). To test for responses to a brief separation from familiar group members, twelve socially-housed animals were placed alone in the novel environment for 20 min, then returned to their social groups ("SEP/RETURNED"; five males, seven females; median age 22.4 months, range 18.0–45.4 months). To test for responses to brief separation from familiar group members in the presence of an unfamiliar opposite-sex animal, six males and six females were placed as pairs into a novel environment for 20 min and remained paired with this partner in a novel standard cage located in an unfamiliar colony room for 24 hr ("SEP/PAIRED"; median age 28.2 months, range 13.0–85.4 months). To test for a response to brief removal from a familiar environment, four singly-housed animals were removed from their home cages, placed alone in the novel environment for 20 min and returned to single housing in their home cages ("SEP/SINGLE"; three males, one female; median age 36.1 months, range 20.4–87.1 months).

Test Procedure

The testing of each group lasted 3 days. On Day 1, baseline behavior observations and blood samples were taken the day prior to the separation test (see below). The separation test occurred for animals in all groups between 0930 and 1330 h on Day 2. Animals were removed from their familiar home cages, and sequentially placed into a novel environment for the 20 min test session. Each animal was removed gently by nylon net from its home cage and carried to a "novel environment" (a 0.65 × 0.65 × 1.0 m wire mesh cage, covered on three sides by opaque Plexiglas) located in an unfamiliar colony room, due to the unavailability of an isolated test room. The novel environment test cage, which was within 1.5 m of the home cage of unfamiliar animals, allowed visual, olfactory, and auditory communication with unfamiliar marmosets. Animals that were paired in the SEP/PAIRED group were placed into the cage at the same time. A test session duration of 20 min was chosen because it was the longest time which allowed us to test all animals in a group in 1 day and still minimize the effects of diurnal hormone variations on cortisol concentrations. After the test session, each animal was gently removed from the test cage, a blood sample was collected, and the marmoset was either returned to its home cage or placed into alternate housing arrangements as described above. Animals in the SEPARATED group were placed in standard cages in an unfamiliar colony room housing the SEPARATED subjects only, which was only available at the time of testing this group. Additional behavior observations and blood samples were taken on Day 3 following the separation test (see below).

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Behavior Observations

The frequency of occurrence of phee calls, which are loud, tonal whistles of approximately 5.5–10.5 kHz and 20 ms–2 sec in duration [Winter, 1978], was determined by ear by an experienced human listener (first author) sitting approximately 0.75 m from the cage. Call frequency was noted using the Observer 3.0® event recording software (Noldus Information Technology, Waginingen, the Netherlands) on a laptop computer (Tandy, model 1800, Fort Worth, Texas). During the 20 min novel environment tests, the tabulation of phee calls began immediately after placement into the novel environment. In the case of paired animals, phees were scored individually for each animal. On Day 1 (pre-separation) and Day 3 (post-separation), the frequency of phee calls was scored in focal animal 20 min home cage observations of marmosets in the SEPARATED, SEP/PAIRED and SEP/RETURNED groups. Pre- and post-separation observations occurred between 1230 and 1530 hr, at least 2 hr after blood sampling procedures were concluded. The SEP/SINGLE animals were not observed in the home cage.

Blood Sampling Procedure

Approximately 0.3 ml blood was drawn from the femoral vein of each animal at four sample times. In the common marmoset, plasma cortisol levels of marmosets sampled at 0900 h and again at 1145 h are statistically indistinguishable [Saltzman et al., 1996]. A "baseline" sample was drawn on Day 1, the day prior to the novel environment test, between 0840 and 1040 hr. Two samples were drawn on Day 2 and one on Day 3 to investigate if any cortisol changes were transient or sustained following either brief or prolonged separation. Day 2 samples were drawn immediately following the novel environment test ("30 min," 0950–1330 hr), and 90 min following the start of the novel environment test ("90 min," 1100–1300 hr). A final sample ("24 hr") was drawn between 0830 and 1040 hr on Day 3 of the study. To minimize the effects of diurnal hormone variations, 24 hr samples were drawn within 10 min of each animal's baseline sampling time. On Day 2, testing procedures necessitated drawing samples across large time spans, but care was taken to sample all treatment groups across similar times of day. An analysis of the mean collection times for each group showed that, on Day 1, the mean time-of-sample of the smallest group (SEP/SINGLE) was earlier than all other groups ($F_{3,31}$=11.845, P<0.0001; Scheffe P's <0.001), and earlier than the SEP/RETURNED and SEPARATED groups (Scheffe P's <0.05) on Day 3. On Day 2, the treatment groups' mean time-of-sample were not significantly different at either 30 min or 90 min. The order of sampling was consistent between days. Animals were manually restrained during blood sampling, which occurred within 3 min of capture. The samples were immediately placed on ice. Samples were centrifuged at 3,000 rpm for 20 min, after which the serum was pipetted off and frozen at –70° C for hormonal assay.

Hormone Assays

Total serum cortisol (CORT) levels were assayed by Assay Services at the Wisconsin Regional Primate Center (Madison, WI). The sensitivity of the cortisol radioimmunoassay is 1.0 μg/dL, and the inter- and intra-assay c.v.'s are 6.7% and 5.3%, respectively [see Saltzman et al., 1994].

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Statistical Analyses

All analyses were performed using statistical programs for Macintosh computers (Statview 4.01 or Superanova 1.11, Abacus Concepts, Inc., Berkeley, CA). The rate of phee vocalizations was tested across all observations by group and sex, by using nonparametric Friedman's test. We used paired Wilcoxon Signed Ranks tests to examine call rates in each group, across pairs of observations, Mann-Whitney U tests to examine call rates between sexes, and Kruskal-Wallis tests to test between groups, for each observation. Cortisol values were examined two ways. First, we used within-subjects repeated measures ANOVAs with post-hoc contrasts (mean comparisons, Fishers PLSD, or Scheffe) to separately test each of the four groups for changes in cortisol values across time. Second, we tested for differences between the groups' cortisol responses. Preliminary analysis revealed that the mean baseline cortisol value of the SEPARATED group, which consisted of males, was significantly different from the other groups ($F_{3,17}$=4.673, P=0.015), being lower than that of males in the SEP/PAIRED (Fishers PLSD, P<0.05) and SEP/SINGLE groups (Fishers PLSD, P<0.05), although not lower than males in the SEP/RETURNED groups. No significant differences were found between groups in female baseline CORT values ($F_{2,11}$=1.999, P=0.182). To test between groups, we compensated for baseline differences by first converting each group's raw cortisol value, sampled at the 30 min, 90 min, and 24 hr time points, to that group's percent change from its baseline value [(time point value/baseline value) $\times$ 100], and then transformed the percent change scores using the arcsine transformation for statistical analysis [Sokal & Rohlf, 1969]. The transformed percent change from baseline values were examined for differences among groups and between sexes, first using a two-way repeated measures ANOVA, and then using single-factor ANOVAs and post-hoc tests at each time point. Cortisol data are graphically represented as percent change from baseline for each group for easier visual comparisons among groups. Regression analysis was used to examine relationships between calling rate in the novel environment and untransformed cortisol levels for each group. Statistical significance was set at P<0.05.

RESULTS

Vocal Response To Separation

The rate of phee calls produced during the novel environment test did not vary by group (Fig. 1; Kruskal-Wallis H=3.37, P=.19). Marmosets in the SEPARATED, SEP/PAIRED, and SEP/RETURNED groups produced more phee calls during the novel environment test than during the pre- or post-test observations (Friedman's χ^2=41.89, P<0.0001); the phee call rate by SEP/SINGLE animals was not measured during Day 1 and Day 3 observations. There were no differences in phee calling rates between SEPARATED, SEP/PAIRED, and SEP/RETURNED groups on Day 1 or Day 3 (Day 1, Kruskal-Wallis H= .269, DF= 2, P= 0.50; Day 2, K-W H= 1.579, DF= 2, P= 0.25). Phee rates by animals in the SEP/PAIRED group were higher on Day 3 than Day 1 (Wilcoxon z=-2.04, P=.041).

There were no sex differences in phee calling rates within the SEP/PAIRED or SEP/RETURNED groups in any observation (SEP/PAIRED, Day 1, Mann-Whitney U= 15.0, P = 0.32, novel environment, U= 13.5, P = 0.47, Day 3, U= 11.0, P = 0.21; SEP/RETURNED, Day 1, Mann-Whitney U= 12.5, P = 0.99, novel environment, U= 15.0, P = 0.68, Day 3, U= 10.0, P = 0.32. Within each sex, neither male nor female phee calling rates differed among treatment groups on

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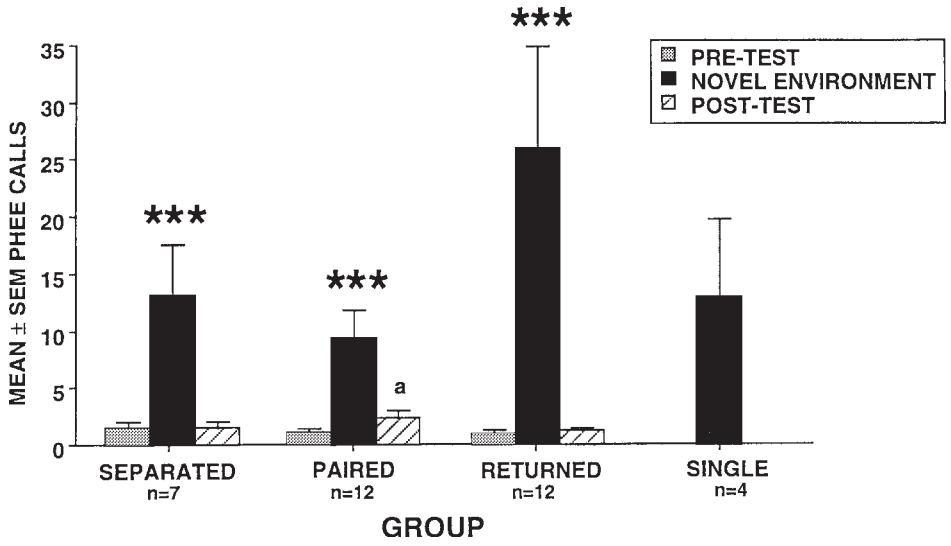


Fig. 1. Mean frequency of phee calls produced during each observation, by marmosets in each group. Animals called more frequently during the novel environment test than in the pre-test or in the post-test observation (***P<0.0001). In the paired group, "a" indicates post-test calling rate is greater than pre-test calling rate (P=0.041).

Day 1 (male, Kruskal-Wallis H= 1.69, DF= 2, P= 0.41; female, Mann-Whitney U= 15.0, P= 0.99), Day 3 (male, Kruskal-Wallis H= .516, DF= 2, P= 0.74; female, Mann-Whitney U= 5.0, P= 0.07), or during the novel environment test (male, Kruskal-Wallis H= 1.99, DF= 3, P= 0.57; female, K-W H= 3.73, DF=2, P= 0.15).

Cortisol Response To Separation

As shown in Figure 2, CORT responses to the test varied by group, where CORT increased in the two groups which experienced prolonged separation from familiar group members. Within-subjects repeated measures ANOVAs revealed that SEPARATED values rose following the novel environment test ($F_{3,18}$=23.939, P=0.0001; baseline, 102.8 ± 9.9 µg/dl; 30 min, 128.3 ± 8.0 µg/dl; 90 min, 175.8 ± 11.0 µg/dl; 24 hr, 287.4 ± 30.0 µg/dl); by 90 min after the separation, CORT was higher than either baseline or 30 min values and was even higher 24 hr later (post-hoc contrasts, P's < 0.05). Cortisol values of the SEP/PAIRED animals also rose following the novel environment test ($F_{3,33}$=5.565, P=0.003; baseline, 219.3 ± 27.2 µg/dl; 30 min, 178.7 ± 18.7 µg/dl; 90 min, 219.4 ± 25.9 µg/dl; 24 h, 309.0 ± 46.5 µg/dl), being higher at 24 hr than at baseline, 30 min, or 90 min (post-hoc contrasts, P's < 0.05). In contrast, cortisol values of the SEP/RETURNED group remained similar across samples (baseline, 186.1 ± 26.3 µg/dl; 30 min, 198.0 ± 18.2 µg/dl; 90 min, 217.1 ± 19.1 µg/dl; 24 hr, 199.2 ± 17.4 µg/dl). SEP/SINGLE values dropped after the novel environment test (baseline, 288.3 ± 49.6 µg/dl; 30 min, 241.2 ± 41.0 µg/dl; 90 min, 233.6 ± 44.0 µg/dl; 24 h, 261.0 ± 72.8 µg/dl); CORT was lower at 90 min than at baseline (post-hoc contrast, P < 0.05).

A two-way repeated measures ANOVA using transformed cortisol percent change values found significant effects of group ($F_{3,31}$=6.81, P <.002) and cortisol percent change ($F_{2,62}$=20.84, P <0.0001), as well as a group and cortisol percent change interaction ($F_{6,62}$=7.43, P<0.0001; see Fig. 2). Breaking down the analysis by time point, one-way ANOVAs showed main effects of group at 90 min

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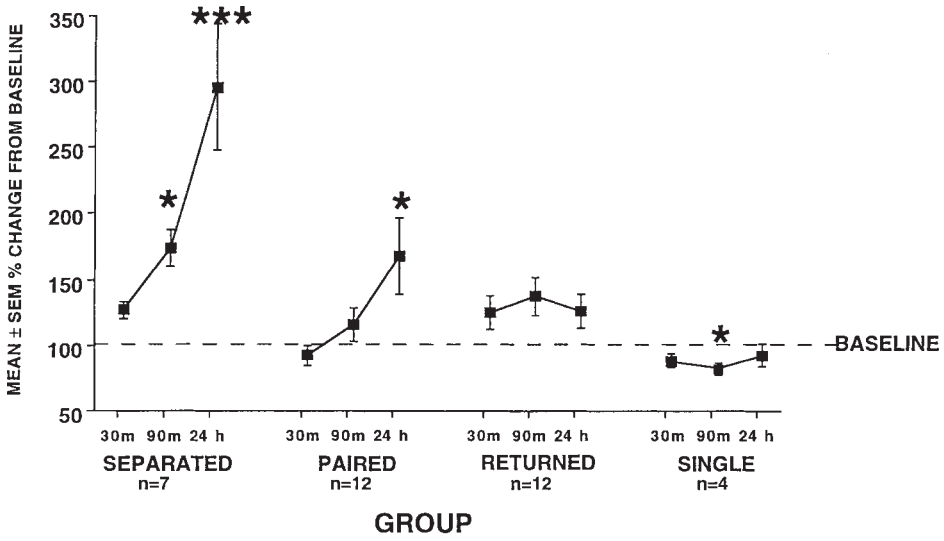


Fig. 2. Cortisol (percent change from baseline) by group. Asterisks indicate significant differences from baseline within each group. See Results for further details. ***P<0.0001 **P<0.001 *P<0.05.

($F_{3,31}$=3.099, P=0.045) and at 24 hr ($F_{3,31}$=5.990, P=0.003), but not at 30 min ($F_{3,31}$=2.177, P=0.116). Post-hoc analyses found that animals which experienced separation from familiar conspecifics (SEP/PAIRED) but were in the presence of a potential new mate, and animals which simply experienced removal from a familiar environment (SEP/SINGLE), showed a smaller percent change at 30 min than the two groups which experienced separation from social partners and were alone in the test arena (SEPARATED and SEP/RETURNED; Fishers PLSD, P's < 0.05). At 90 min, SEP/SINGLE animals had a smaller percent change than SEP/RETURNED animals (Fishers PLSD, P < 0.05). The percent change by animals that remained separated without conspecifics (SEPARATED) was greater than all other groups at both 90 min and 24 hr (Fishers PLSD, P's < 0.05).

To test for a sampling order effect, CORT was tested as a function of the order at each sample time, using regression analyses. There was no significant relationship between sampling order and CORT value. There were no main effects of sex or age in the SEPARATED, SEP/PAIRED or SEP/RETURNED groups on CORT values at any time point, but the CORT values of the only female in the SEP/SINGLE group were higher than male values at all sample points (one-sample-to-a-mean t-tests, Ps < 0.03). Marmosets with lower baseline CORT had a greater percent change from baseline at all sample times (30 min, $F_{1,33}$=28.94, P <.0001, r^2=.47; 90 min, $F_{1,33}$=43.66, P <.0001, r^2=.57; 24 hr, $F_{1,33}$=20.61, P <.0001, r^2=.38).

There were no significant relationships between cortisol levels and the total number of phee calls produced in the novel environment in any group.

DISCUSSION

Marmosets called most frequently during the novel environment test, before cortisol levels increased, and called infrequently 24 hr after separation, when cortisol was most elevated. These results extend to the common marmoset the finding of a temporal disassociation between behavioral and physiological re-

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sponses to involuntary separation [e.g., Coe et al., 1978; Hennessy et al., 1979; Bayart et al., 1990; Wiener et al., 1990; Levine et al., 1993]. Infant distress vocalizations are an adaptive response to reestablish contact with the mother [Rosenblum & Plimpton, 1981]. Cortisol levels may have continued to rise in the groups experiencing prolonged separation because the initial behavioral response, the production of phee calls, was insufficient to obtain reunion with conspecifics in our paradigm. The eventual suppression of separation vocalizations may be adaptive, since a quiet animal would conserve energy and be less likely to attract predators [Hennessy, 1997]. Studies of guinea pig pups separated from their mother suggest that elevated CRF levels act to suppress separation vocalizations [Hennessy et al., 1997]. The particularily high levels of cortisol, and absence of vocalizations observed in our subjects separated for 24 hr from familiar conspecifics, suggests that a similar mechanism may act in adults of a socially-bonding nonhuman primate.

One of the purposes of this study was to contribute to an understanding of the early stages of pair formation. Information on the early stages of pair formation in wild marmosets is not available. We do not know the time course over which reproductively mature individuals, who are ready to leave their natal group, search for and select a mate or whether this process involves a prolonged period of evaluating several potential mates before one is selected. Mate selection may occur relatively rapidly, given the greater dangers from predation that can result from living alone. Our procedure of pairings in the SEP/PAIRED subjects eliminated the possibility of mate choice. Nevertheless, it is instructive that all of the pairings eventually resulted in the production of live offspring. Therefore, we believe that the findings in this study, pertaining to the influence of an unfamiliar potential partner on the separation response, have relevance to the process of mate selection in the wild. As our study showed, the production of phee calls stimulated by exposing adult common marmosets to a novel environment is neither attenuated nor increased by the presence of an unfamiliar opposite-sex marmoset in the first 20 min after pairing, as might be expected if unfamiliar marmosets were perceived to be a threatening or distressful stimulus. However, newly-paired marmosets produced more phee calls at 24 hr than they did prior to pairing, in contrast to the other test groups. Since phee calls can be produced by marmosets in an intergroup context as well as separation context [Epple, 1968; Pook, 1978; Hubrecht, 1985; Norcross & Newman, 1993; Norcross et al., 1994], and we have found that non-breeding marmosets separated from their natal groups produce intergroup phee calls four days after pairing [Norcross & Newman, 1997], it is possible that these calls represent an early response to pair formation.

The calling rate of the singly-housed animals was not significantly different from that of their socially-housed counterparts during the novel environment test. The relevant component(s) of the home environment, whether animate or inanimate, that elicited the separated marmosets' behavioral responses remains undetermined. Since the singly-caged animals were normally housed within a room containing other marmosets, they may have regarded neighboring marmosets within visual, auditory, and olfactory contact as part of their social group. Alternatively, singly-housed adult marmosets may, like other species [Gunnar et al., 1980; Coe et al., 1982; Hennessy, 1984; Levine & Wiener, 1988], be stimulated to call simply by the separation from a familiar environment; attachment bonds can be formed with inanimate objects [Cairns, 1966]. These individuals, along with the other subjects, also may have been stimulated to call by the presence of unfamiliar marmosets in close proximity, or the novel environment. Although separation calls are generally assumed to function to reunite the separated

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animal with a familiar attachment object, separation phee calls by adults may also function to unite emigrating marmosets with potential pairmates.

Prolonged separation, such as might be found following death of a social partner, appears sufficient to activate the HPA axis, as shown by the elevated cortisol levels seen in the SEPARATED and SEP/PAIRED groups. However, the cortisol response of the newly-paired marmosets appeared attenuated compared to animals separated 24 hr without social partners, even though an unfamiliar social partner did not influence the vocal response to separation. Association with a potential mate may have had stress-buffering effects. The cortisol levels of the paired animals did, however, persist above baseline levels at 24 hr. In comparison, Smith and French [1997] found that, in Wied's black tufted marmosets separated for 11 hr from their normal social groups, post-separation urinary cortisol was significantly elevated but returned to baseline by the morning after reunion. Whether the persistently elevated cortisol found in separated marmosets in our study results from prolonged separation from familiar social groups, prolonged exposure to novelty, or social instability remains to be determined. Glick [1984] found that 20 min of exposure to unfamiliar females elevated male cynomolgus macaques' cortisol levels, particularly when the exposure occurred in an unfamiliar cage rather than in the home cage in the presence of familiar peers. Since the paired animals' cortisol at 24 hr was lower than that of the group separated without social partners at 24 hr, it seems unlikely that cortisol rose as a subsequent consequence of exposure to an unfamiliar animal [e.g., Glick, 1984]. However, both the prolonged absence of social partners, and the instability of social relationships appears to stimulate HPA activation in the common marmoset. Peer group formation can elevate cortisol in marmosets to levels indistinguishable from cortisol levels of the same animals in isolation, which in turn is significantly higher than the cortisol levels of the same animals in stable peer groups or in heterosexual pairs [Johnson et al., 1996]. However, Saltzman et al. [1994] found that elevated cortisol levels seen in marmoset females at 24 and 48 hr after peer group formation were less clearly related to changes from pair to peer housing than with ovarian activity and physical wounding.

Cortisol levels sampled immediately after the novel environment test were not significantly elevated in any group. This suggests that the 20 min novel environment test, although eliciting a behavioral response to separation from familiar conspecifics, may not have been sufficiently arousing to stimulate increased HPA activity. Similarly, Saltzman et al. [1996] found that marmosets which had been habituated to a separation test environment did not show elevated cortisol levels when sampled immediately after a 25 min separation from either a heterosexual partner or a mixed sex group. In our study, the lack of an immediate cortisol response was surprising, because the separation test occurred in an unfamiliar environment in view of unfamiliar animals. However, since our marmosets are accustomed to handling and biweekly cage transfers, the characteristics of the brief novel environment test may have been relatively benign.

Care was taken to control for normal diurnal variation in cortisol levels by testing each group at similar times of day. However, differences were found between some groups in baseline cortisol, which cannot be explained by pre-test housing conditions, age of animals, time of sampling, or inter-assay variability. In diurnal species, cortisol levels are generally highest prior to or upon waking, gradually decline throughout the day, and reach a nadir shortly before or at retiring to sleep [e.g., Coe & Levine, 1995; Lyons et al., 1995]. To control for variances between groups, we examined each group's absolute cortisol values in a repeated-measures design, and, when testing between groups, compared the

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groups' percent change from baseline, rather than absolute cortisol values. However, in comparisons between groups, the magnitude of the cortisol response must be interpreted carefully. Because the magnitude of cortisol responses to stressors can be inversely related to baseline cortisol values [e.g., Coe & Levine, 1995; Smith & French, 1997; this study], the largest percent change seen in the SEPARATED group may be confounded by this group's lower absolute baseline cortisol. In contrast to the cortisol elevations found in animals undergoing prolonged separation, the lack of any cortisol response in animals separated only 20 min from their social group or home cage (SEP/RETURNED, SEP/SINGLE) supports the presence of stress-buffering properties in familiar social and/or physical environments. Prolonged separation of singly-housed animals from familiar environments would be necessary to determine if a separation response can result from the length of exposure to a novel environment.

Singly-housed marmosets showed a modest decline from baseline cortisol levels at 90 min after the separation test. Although this may reflect the decline in cortisol normally seen in the diurnal rhythm of primates, a reduction in cortisol can be linked to either the occurrence or the anticipation of positive events [see Mendoza & Mason, 1986]. Mendoza & Mason [1986] suggest that the initial decline in cortisol of heterosexually-housed squirrel monkey females presented with an unfamiliar female is in response to the potential alleviation of social deprivation. Similarly, singly-housed marmosets may be undergoing social deprivation, which the unfamiliar marmosets in the test colony room have the potential to alleviate. The higher mean baseline CORT value of the singly-caged group suggests that any response by these marmosets may have been limited by chronically higher absolute cortisol values. However, these baseline values were not significantly higher than animals in either the socially-housed SEP/PAIRED or SEP/RETURNED groups, and fell well within the range of cortisol values found in this study (range 69.8–631.2 μg/dL across conditions).

Different information about an animal's state may be obtained from behavioral compared to physiological measures [Levine et al., 1993]. Although marmosets show an apparent behavioral adaptation to prolonged separation, the robust cortisol response found 24 hr after the separation contradicts rapid adaptation to a new environment. This finding may be relevant to husbandry and research practices involving this species. Longitudinal assessment of a physiological response to separation from familiar group members, both in a familiar and a novel environment, and the subsequent development of the pairbond, remains to be accomplished. Further study of the immediate physiological response of long-term pairbonded marmosets with like- and opposite-sex conspecifics, both alone and in the presence of the pairmate, may clarify mechanisms contributing to the development, maintenance and quality of the pairbond in this species.

CONCLUSIONS

1. All marmosets produced phee calls during the novel environment test, indicating that a 20 min removal from a familiar social partners and/or from a familiar environment is sufficient to elicit a vocal response. The presence of an unfamiliar, opposite-sex conspecific did not affect the call production rate.

2. The marmoset HPA axis is responsive to prolonged changes in surroundings; this response may be influenced by characteristics of the social environment. Cortisol levels were elevated in separated marmosets at 90 min and even higher 24 hr later. When separation coincided with placement with an opposite-sex conspecific, the cortisol response was higher than baseline only after 24 hr of

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cohabitation. A separation interval of 20 min has little affect on the cortisol levels of nonpaired marmosets if they are returned to their previous social group or familiar environment.

3. Although marmosets reliably produced phee calls after placement into the novel environment, call production is inversely related to cortisol levels. Phee calling was highest during the novel environment test, before a cortisol increase was detected. With the exception of animals that remained paired with a partner, animals' phee calling rates returned to pre-separation levels by 24 hr after the test.

ACKNOWLEDGMENTS

We thank D.H. Abbott, C.S. Carter, M. Champoux, R.L. Roberts, and three anonymous reviewers for their helpful comments on an earlier draft of the manuscript.

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